Winding and focussing for geodesics passing a thin cuspidal neck

Jørgen Olsen Lye
(joint work with Daniel Grieser)

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Geodesics

Geodesics are central to geometry. They

- determine the geometry;
- give rise to special coordinates;
- propagate singularities of the wave operator;
- relate to spectral data (wave trace, Selberg trace, etc.);
- are the world lines of free particles in general relativity;
- describe winding states in string theory.

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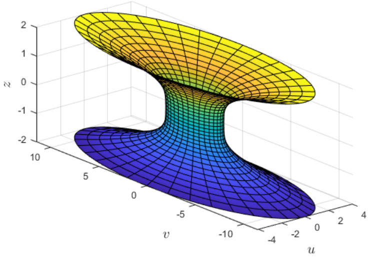
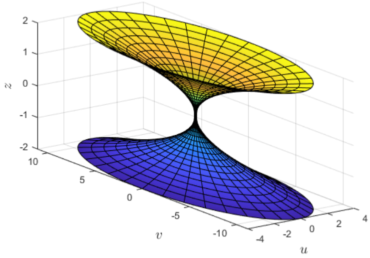
Setting

We study a family of metrics degenerating to a cuspidal singularity.





Jørgen Olsen Lye (joint work with Daniel Grieser)
University of Cologne
Faculty of Mathematics and Natural Sciences

Setup

$$Y \text{ is } (n-1)-\text{dimensional, compact, and smooth;}$$

$$M = [-1,1]_z \times Y;$$

$$\varepsilon \in [0,1];$$

S,b,h are (ε, z)-dependent families of functions, 1-forms and Riemannian metrics on Y, respectively.

$$w \overset{e.g.}{=} \|(\varepsilon, z)\|_p = \sqrt[p]{\varepsilon^p + |z|^p}, \quad p \in [2, \infty).$$

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For $k \geq 2, \kappa \geq k$, consider

$$g = g_\varepsilon = (1 - w^\kappa S)\, dz^2 + 2w^{2k}\, dz \cdot b + w^{2k}\, h\,. \tag{1}$$

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Examples

Let $Y \subset \mathbb{R}^{N-1}$ be a smooth, closed submanifold of dimension $n-1$. Let

$$\Psi_\varepsilon \colon \mathbb{R} \times \mathbb{R}^{N-1} \to \mathbb{R} \times \mathbb{R}^{N-1}$$
$$\Psi_\varepsilon(z, y) = (z, w(\varepsilon, z)^k y)$$

and

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Then the induced metric has the form

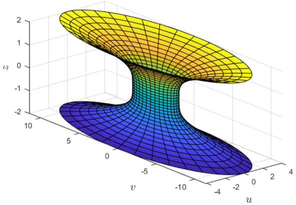
$$g = g_\varepsilon = (1 - w^{2k-2} S)\, dz^2 + 2w^{2k}\, dz \cdot b + w^{2k}\, h \qquad (2)$$

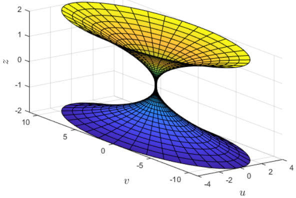
in suitable coordinates ((1) with $\kappa = 2k - 2$). Furthermore,

$$S(\varepsilon, z, y) = a(\varepsilon, z)\, |y|^2 + \mathcal{O}(w^{k-1})$$

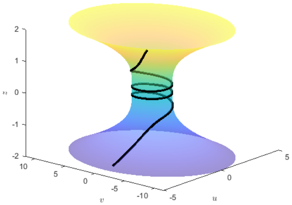
E.g. $Y \subset \mathbb{R}^2$ is an ellipse and $w = \|(\varepsilon, z)\|_{2k}$.

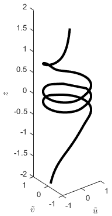
$$M = \left\{ u^2 + \frac{v^2}{1-\delta^2} = z^{2k} + \varepsilon^{2k} \right\} \subset \mathbb{R}^3 \tag{3}$$

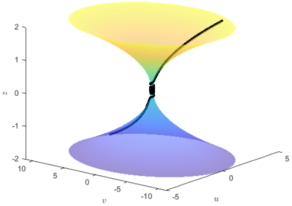


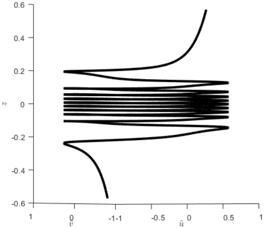


Result 1: Winding









Result 1: Winding

Fix some $T > 0$ and consider geodesics $\gamma(t) = (z(t), y(t))$ starting at $z = 0$ at $t = 0$. Define the horizontal length as

$$\mathrm{length}_Y(\gamma) := \int_0^T |\dot{y}(t)|\, dt.$$

Winding Theorem (Grieser-L. '25)

A geodesic on (M, g_ε) starting at the waist upward at an angle $\varphi \in (0, \frac{\pi}{2})$ to the horizontal has angular length

$$\mathrm{length}_Y(\gamma) \sim \frac{\mathcal{C}_\varphi \cos\varphi}{\varepsilon^{k-1}}. \tag{4}$$

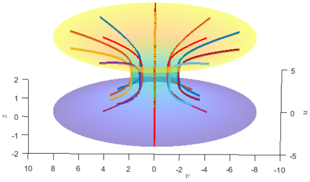
The constant $\mathcal{C}_\varphi$ is decreasing in φ, $\mathcal{C}_\varphi \geq \mathcal{C}_{\pi/2} > 0$, and depends only on φ and the function w. The asymptotics (4) are uniform in φ for compact subsets of $(0, \pi/2)$.

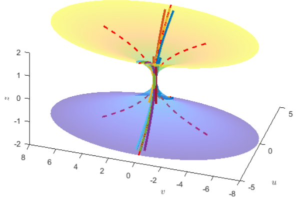
Jørgen Olsen Lye (joint work with Daniel Grieser)

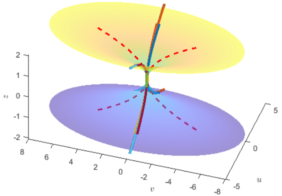
Faculty of Mathematics and Natural Sciences

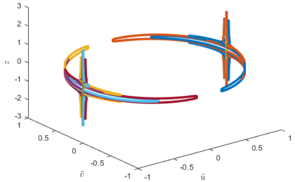
University of Cologne

Result 2: Focussing









Jørgen Olsen Lye (joint work with Daniel Grieser)

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University of Cologne

Result 2: Focussing

$$g = g_\varepsilon = (1 - w^\kappa S)\, dz^2 + 2w^{2k}\, dz \cdot b + w^{2k}\, h\,.$$

Focussing Theorem (Grieser-L. '25)

Assume that $S_0 = S_{|(\varepsilon,z)=0}$ is a Morse function and that $\kappa = 2k - 2$. Consider a family of unit speed geodesics $\gamma_\varepsilon = (z_\varepsilon, y_\varepsilon)$ passing the waist $z = 0$ at $t = 0$, with $(y_\varepsilon(0), \varepsilon\dot{y}_\varepsilon(0))$ converging to some $(y_0, \tilde{v}_0) \in TY$ as $\varepsilon \to 0$. Then for generic limit $(y_0, \tilde{v}_0)$, the point $\gamma_\varepsilon(t)$ will approach $\gamma_{y_{\min}}(t)$ as $\varepsilon \to 0$, for some minimum $y_{\min}$ of S_0, for any fixed $t > 0$.

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Necessity of an assumption on S

When

$$M = \left\{ u^2 + \frac{v^2}{1-\delta^2} = z^{2k} + \varepsilon^{2k} \right\} \subset \mathbb{R}^3,$$

$S_0 = a \cdot |(u,v)|^2$. This is Morse $\iff \delta \in (0,1)$.

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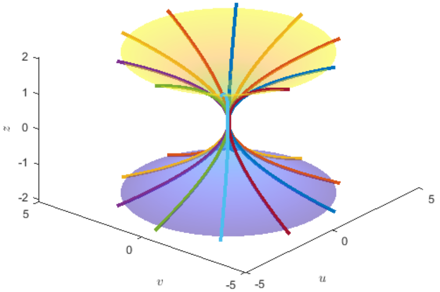

Sketch of proof of Winding I

Start with the warped product case,

$$g = dz^2 + w^{2k} h.$$

If (z, ξ, y, η) are coordinates on $T^*[-1, 1] \times T^* Y$, then

$$2\mathcal{H} = \xi^2 + \frac{|\eta|^2}{w^{2k}}$$

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$$\text{length}_Y(\gamma) = \int_0^T |\dot{y}|\, dt = \int_0^{z(T)} \frac{|\eta|}{w^k \sqrt{w^{2k} - |\eta|^2}}\, dz = \mathcal{O}(\varepsilon^{1-k}).$$

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Sketch of proof of Winding II

In the general case,

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$$\mathrm{length}_Y(\gamma) = \int |\dot{y}|\, dt = \left(\frac{|\eta|}{w^k \sqrt{w^{2k} - |\eta|^2}} (1 + \mathcal{O}(w^\kappa)) + \mathcal{O}(1) \right) dz.$$

The result follows by a careful analysis of this.

Jørgen Olsen Lye (joint work with Daniel Grieser)
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- Blow up at $(\varepsilon, z) = (0, 0)$ and introduce a rescaled cotangent bundle where $\theta = \frac{\eta}{w^{2k-1}}$ is the new Y-momentum variable.

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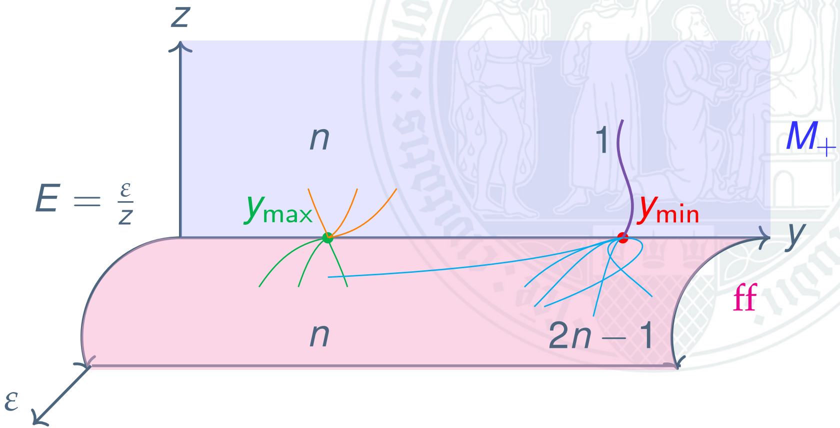
- If $\kappa = 2k - 2$, the geodesic flow becomes smooth on this blown-up and rescaled cotangent bundle.
- Linearise the flow. S_0 Morse implies hyperbolicity. Study the flow using the stable manifold theorem and similar dynamical methods.
- The flow restricted to the front face is a driven Hamiltonian system and can be analyzed.
- Build upon results by Grandjean-Grieser for the flow restricted to $\varepsilon = 0$.

Jørgen Olsen Lye (joint work with Daniel Grieser)

University of Cologne

Faculty of Mathematics and Natural Sciences

Sketch of proof of Focussing II



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